Memory Simulation

The program simulates the functionality of a full memory system, primarily focusing on the page table and TLB. It translates a series of 16-bit logical addresses into physical memory addresses. The virtual memory space contains 256 pages, with pages and physical frames all being 256 bytes.

The user passes in the reference sequence, frame count, and page replacement algorithm when running the program from the command line. The reference sequence file is parsed using our parse_file function, and each logical address is accessed. These addresses are read into an array representing physical memory.

The page table contains one entry per logical page, each of which contain a load bit representing whether that page has been loaded into physical memory, the physical frame number corresponding to the page number, and a 'last-used' timestamp that represents the last time it was accessed. This timestamp is used to implement the LRU PRA.

Physical memory is allocated a variable amount of memory, dependent on the command line argument for frame count. Another array allows frame number to be used as an index to retrieve a corresponding page number, used when evicting a frame and updating the page table accordingly.

When a page fault occurs, we read a page (256 bytes) from BACKING_STORE-1.bin to a free frame, update the page table, and update the TLB.

The FIFO PRA uses a frame index that rotates through the page table. After all frames are filled, the victim is selected from the beginning of the page table. The LRU PRA looks at the 'last-used' timestamp associated with each page table entry, comparing them to see which entry was accessed 'least recently' and selecting it as the victim. The OPT PRA looks at each page table entry and detects which one is accessed farthest from the current position in the reference sequence, then selects it as the victim.

The FIFO PRA has its full functionality implemented within main, but LRU and OPT have dedicated helper functions, each of which return the frame number of the victim frame. The old page associated with the frame to be evicted is accessed through the frame_to_page array. This page then has all associated values reverted to their defaults. Its load bit then becomes 0, indicating it is no longer stored in physical memory. Each TLB entry is checked against the evicted page number to ensure it does not remain present.